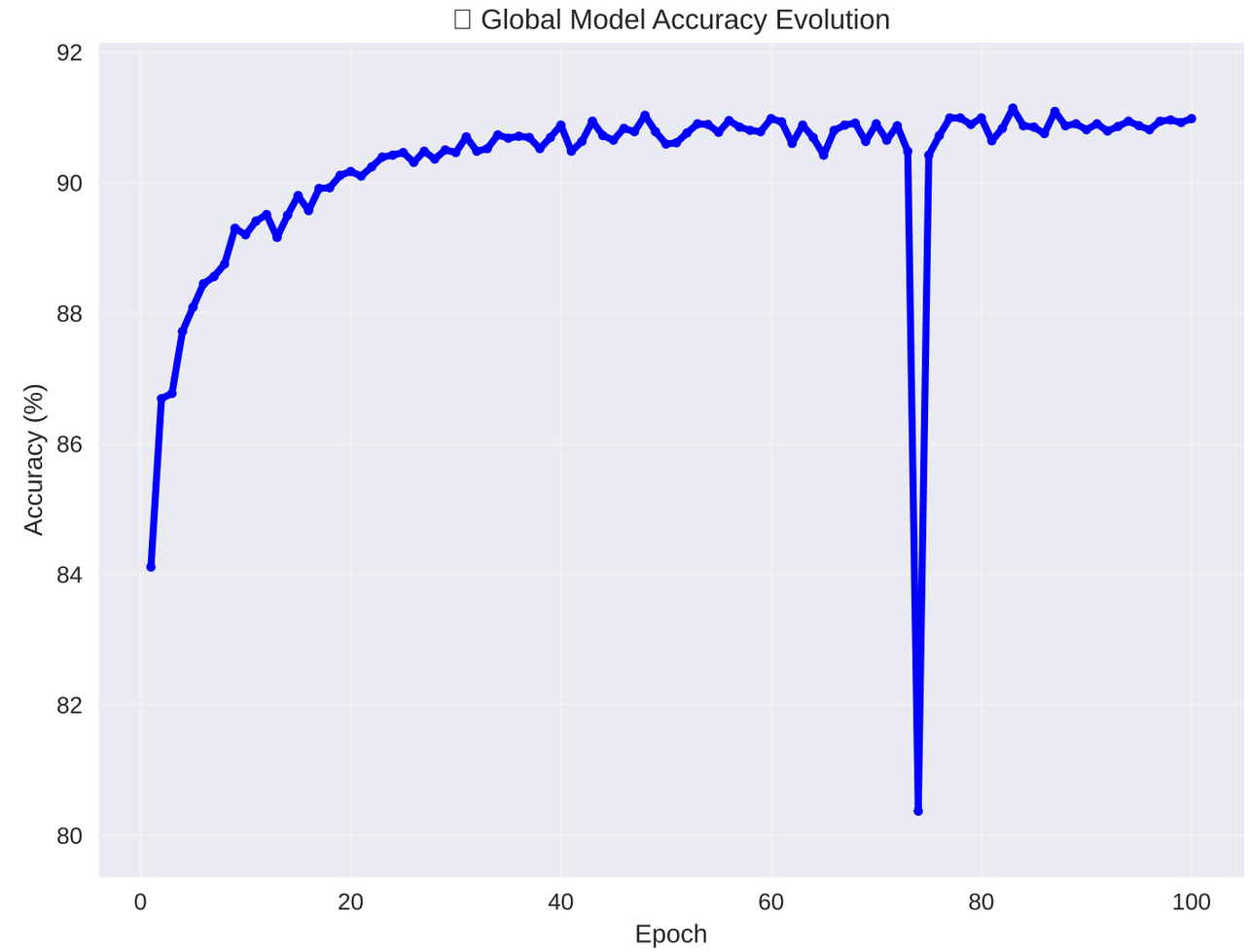
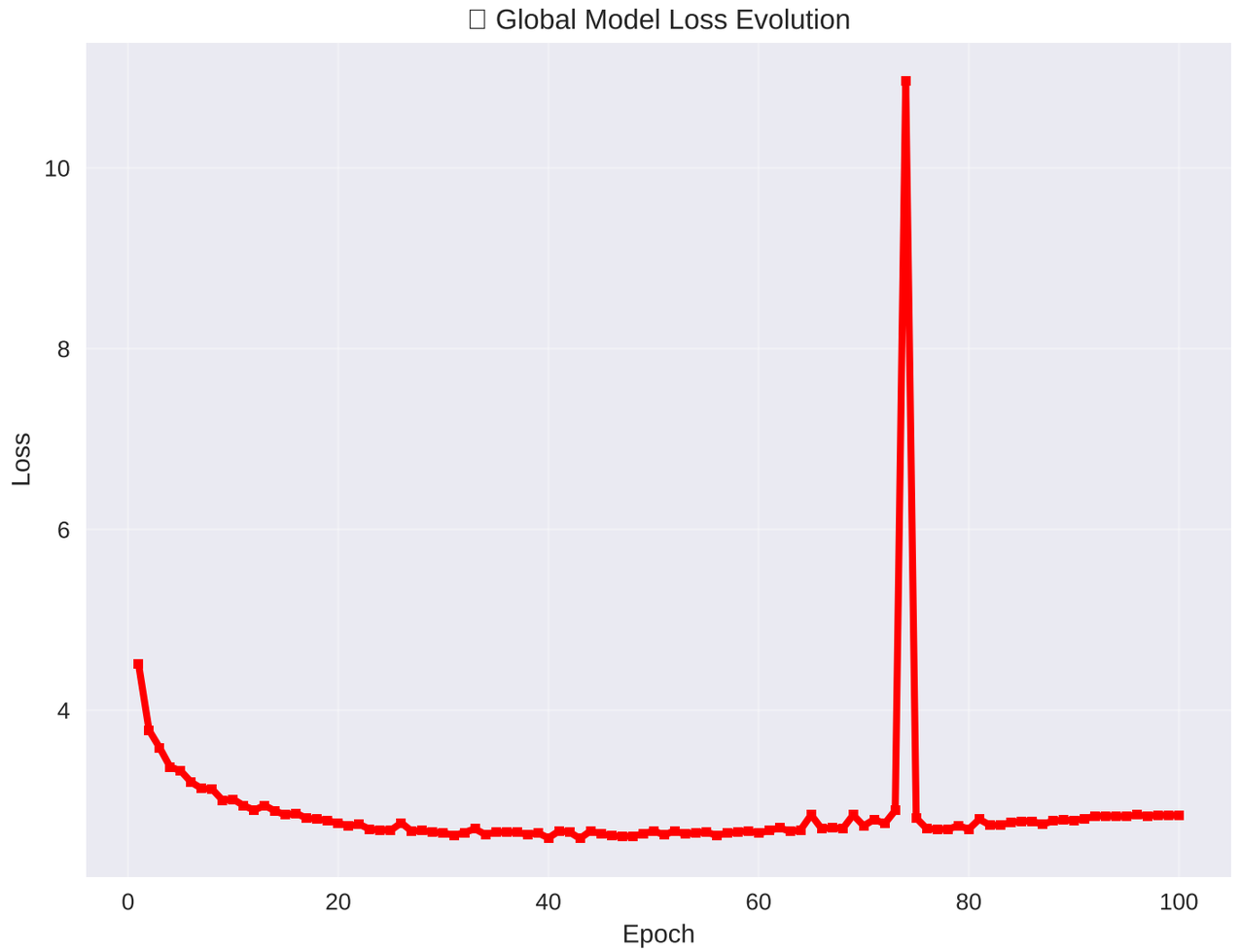
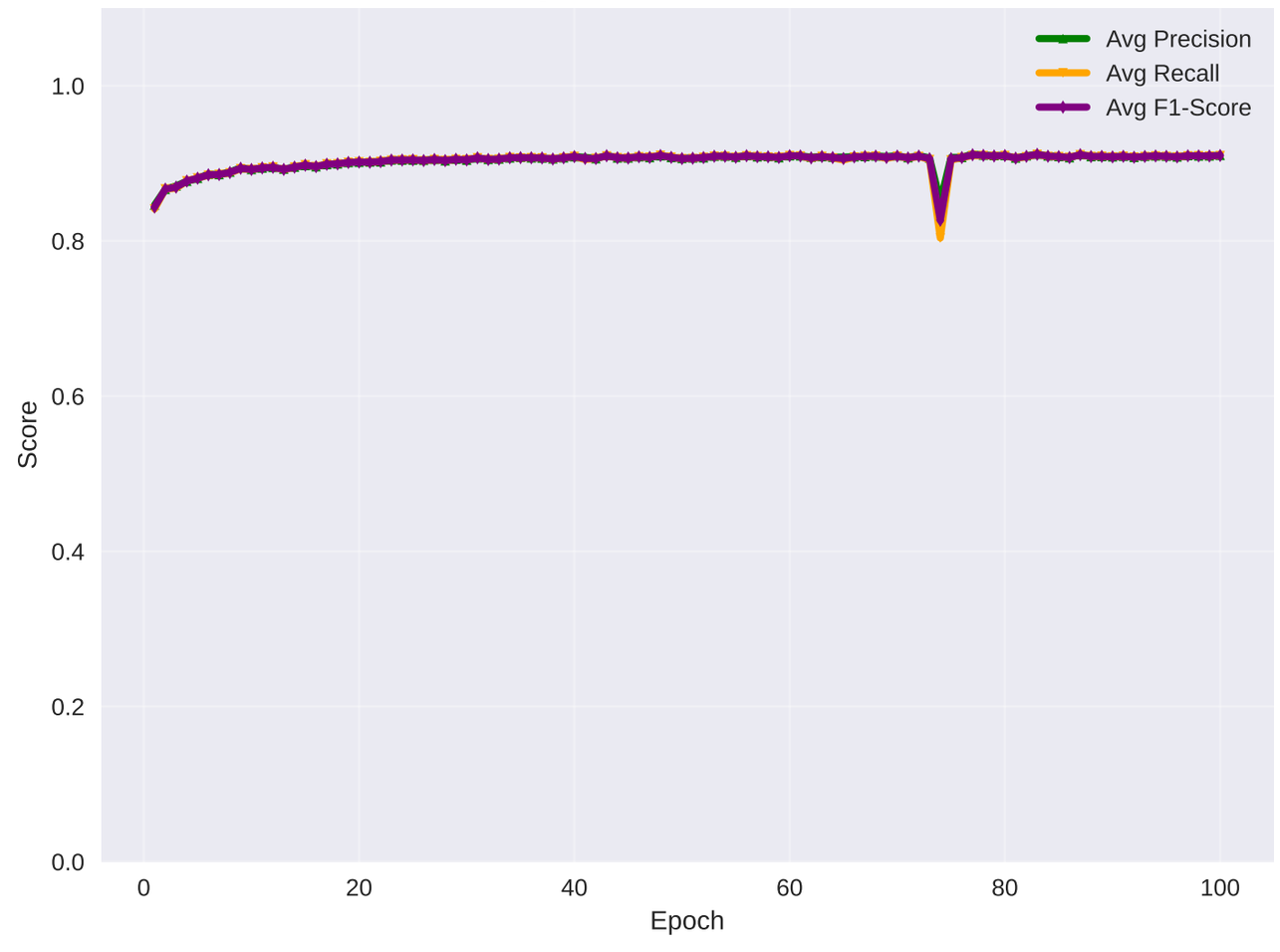
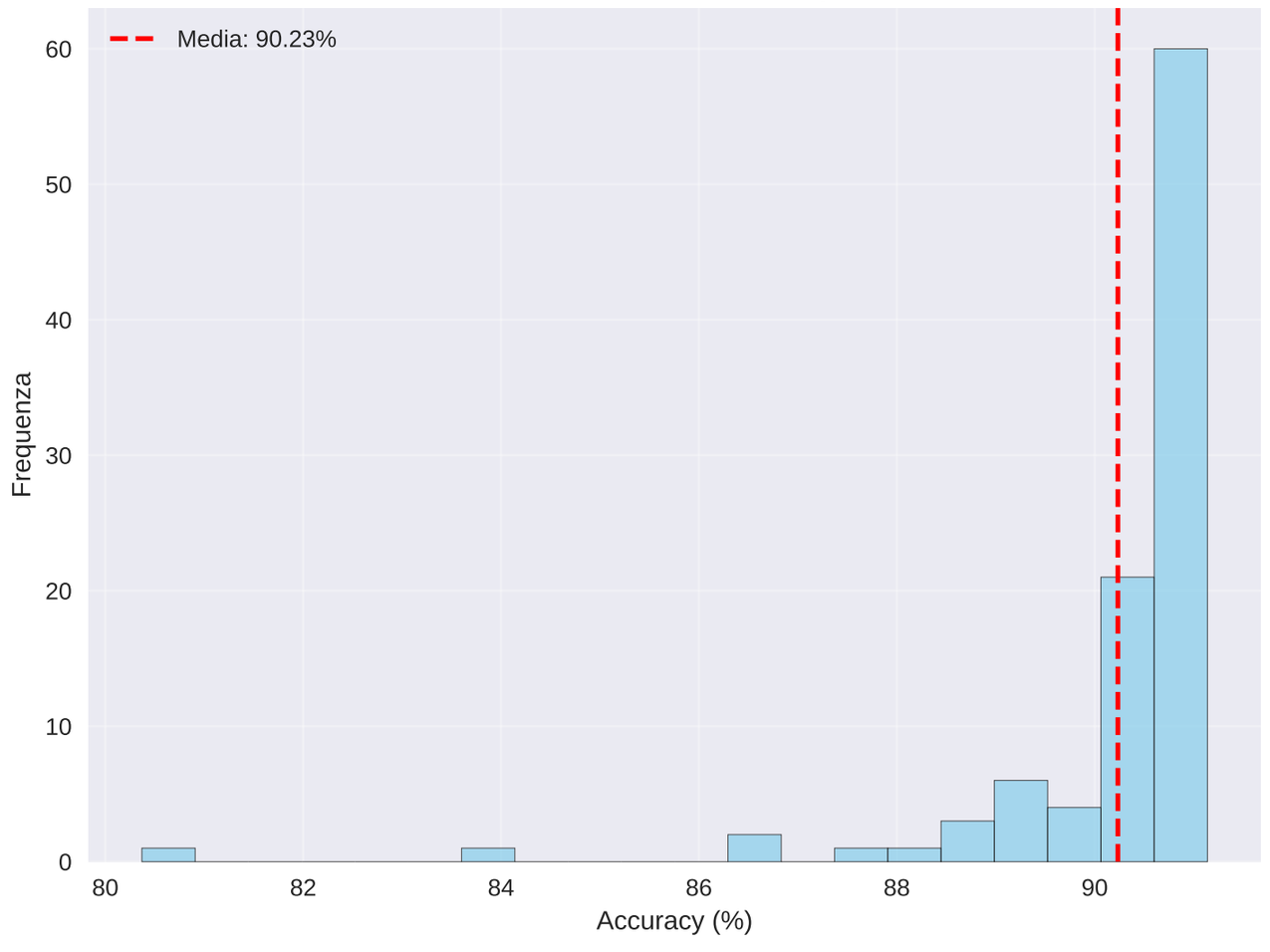
Federated Learning - Main Metrics Overview

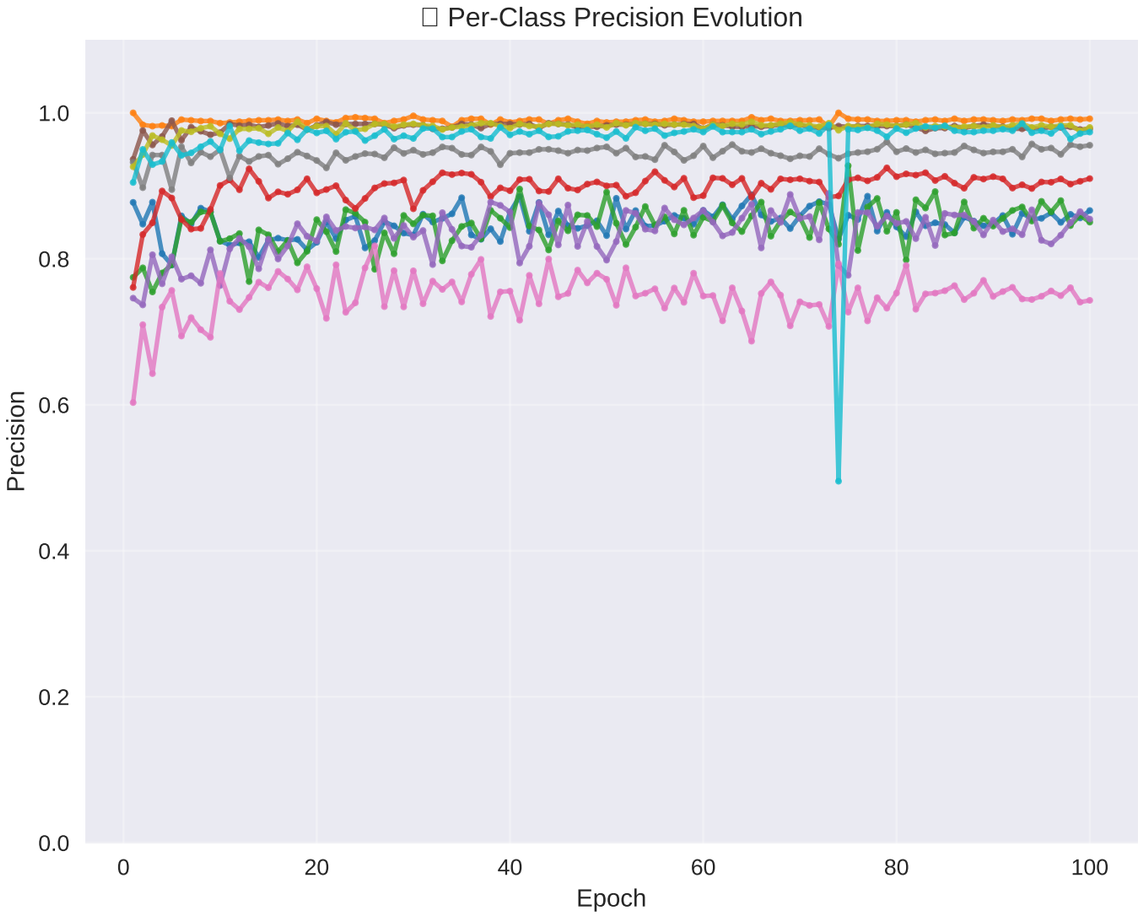




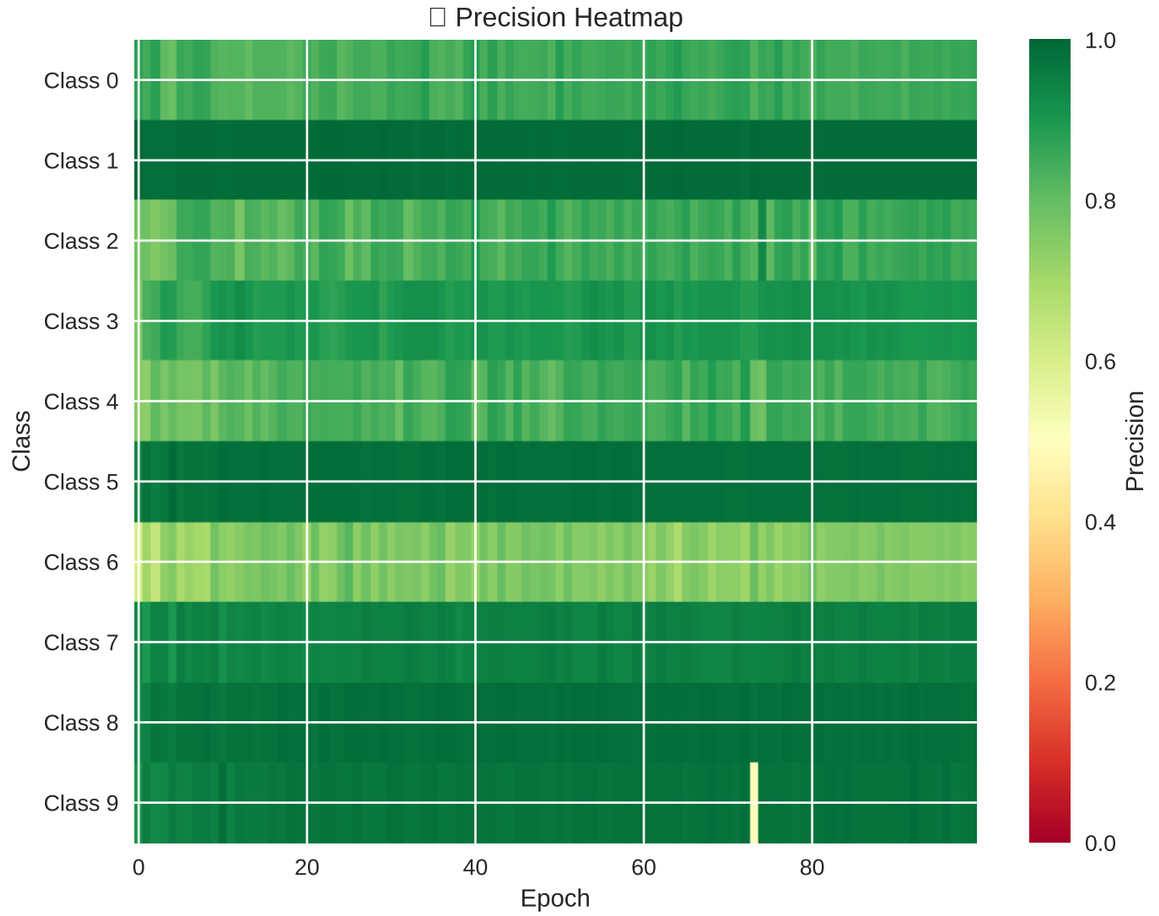


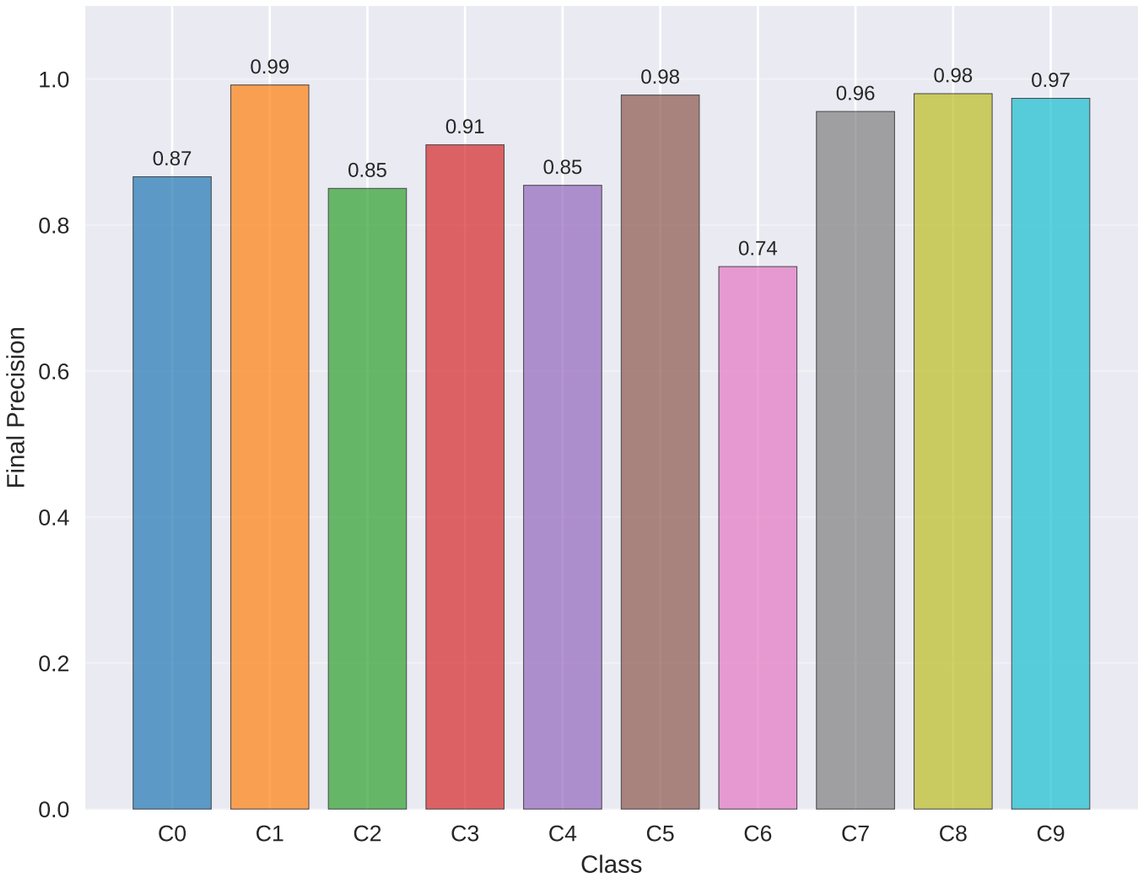


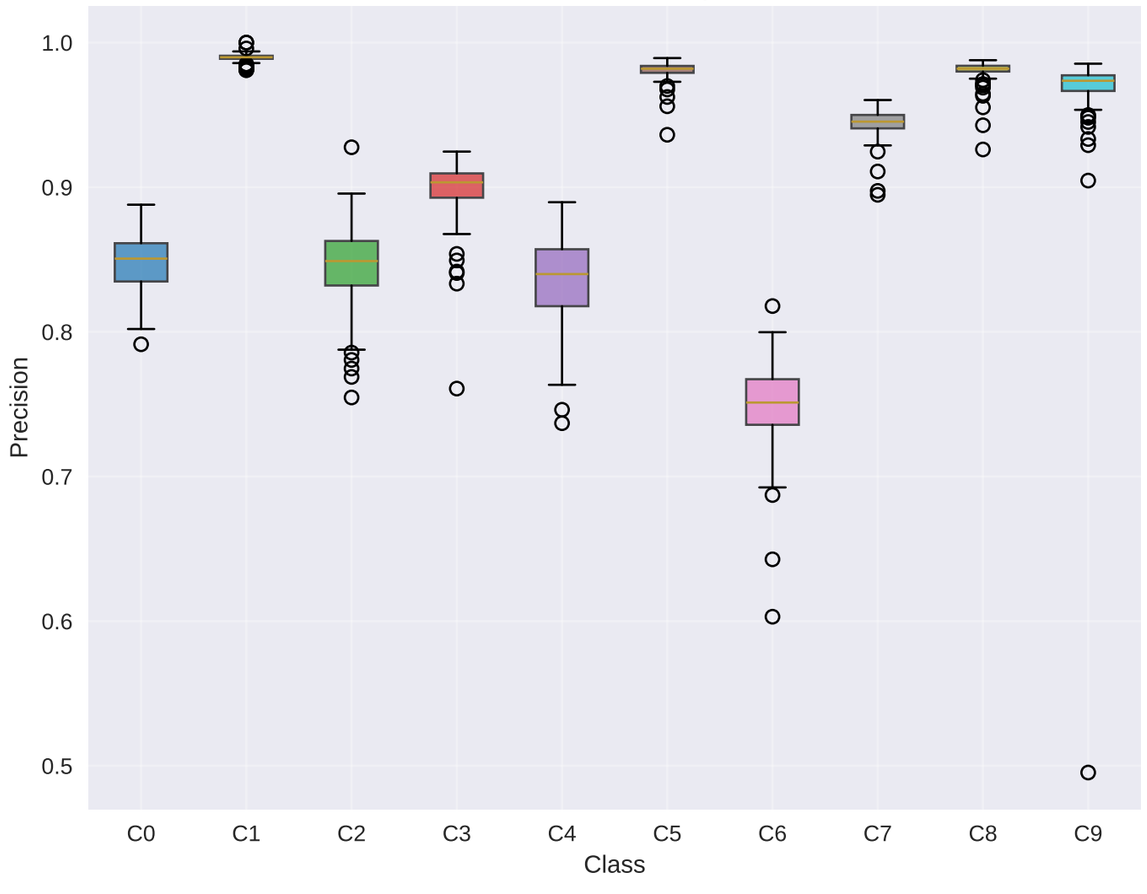
Per-Class Precision Analysis



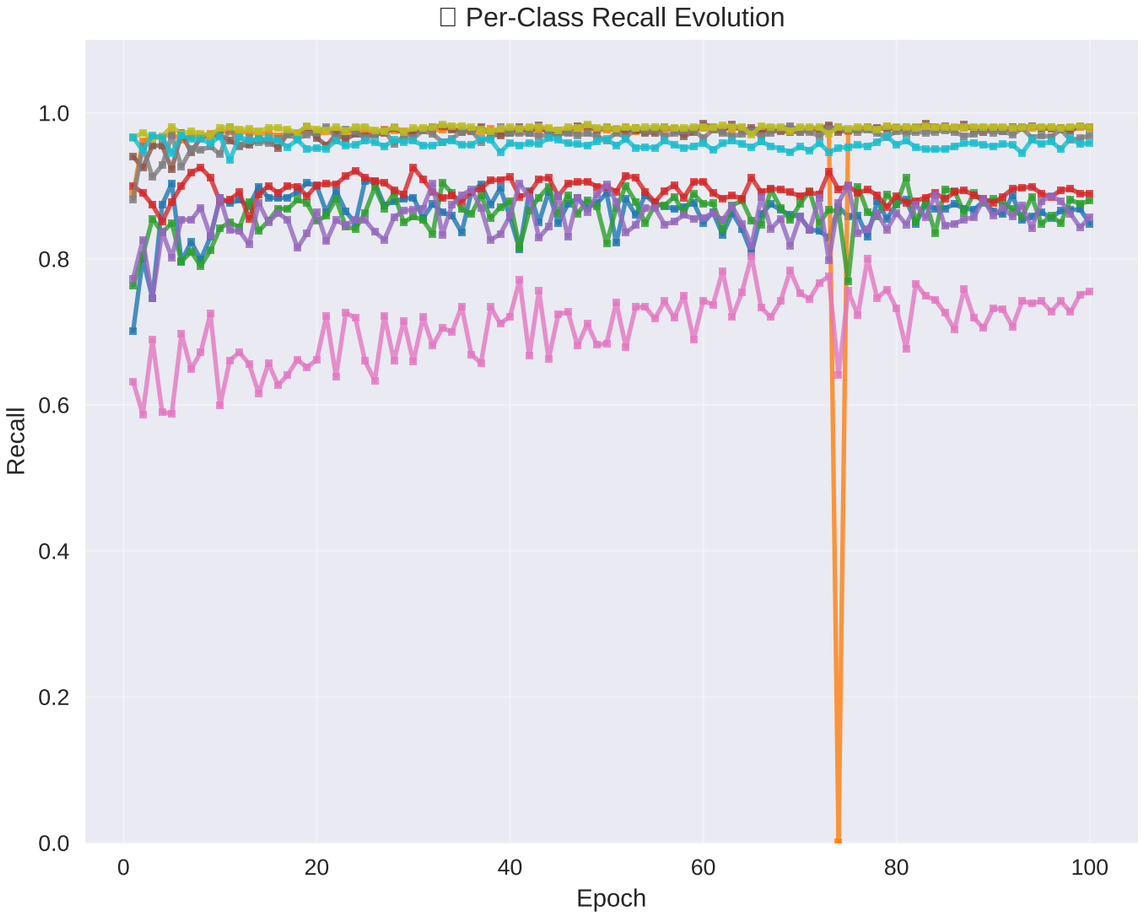


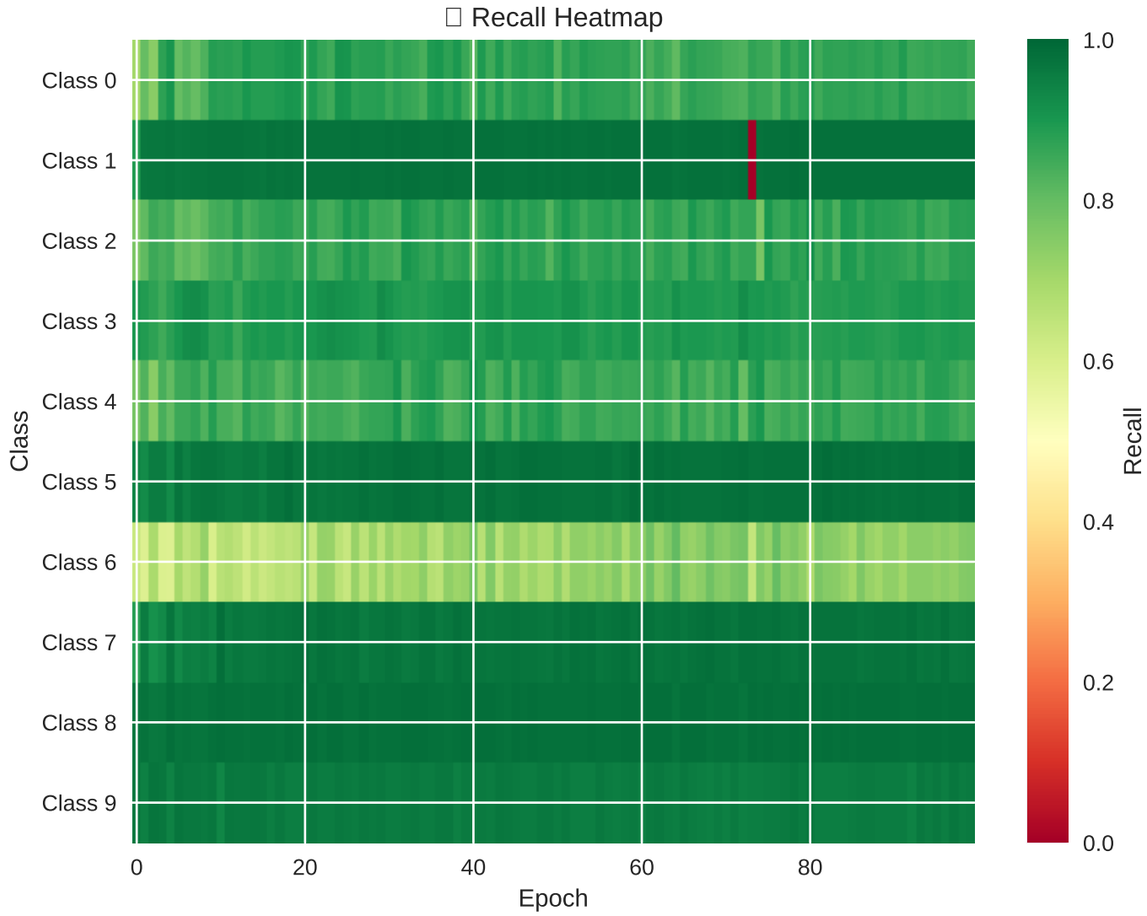


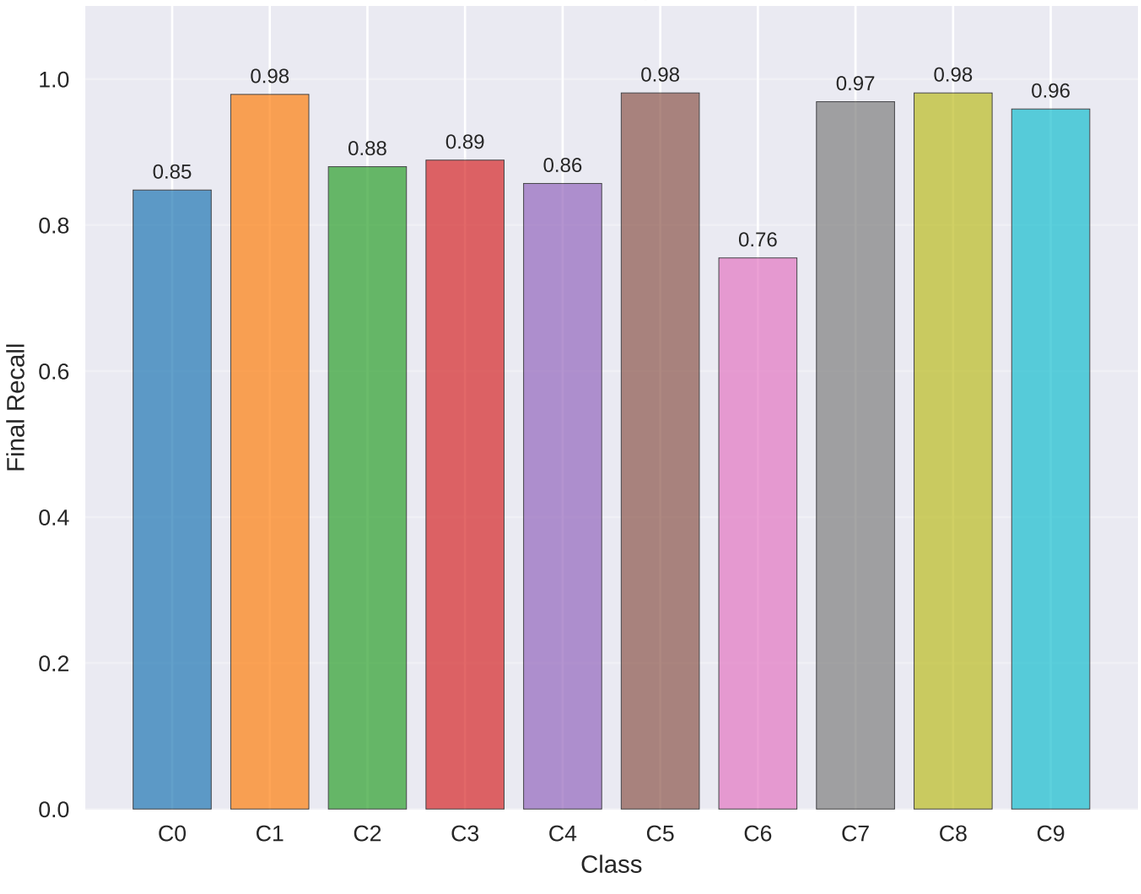


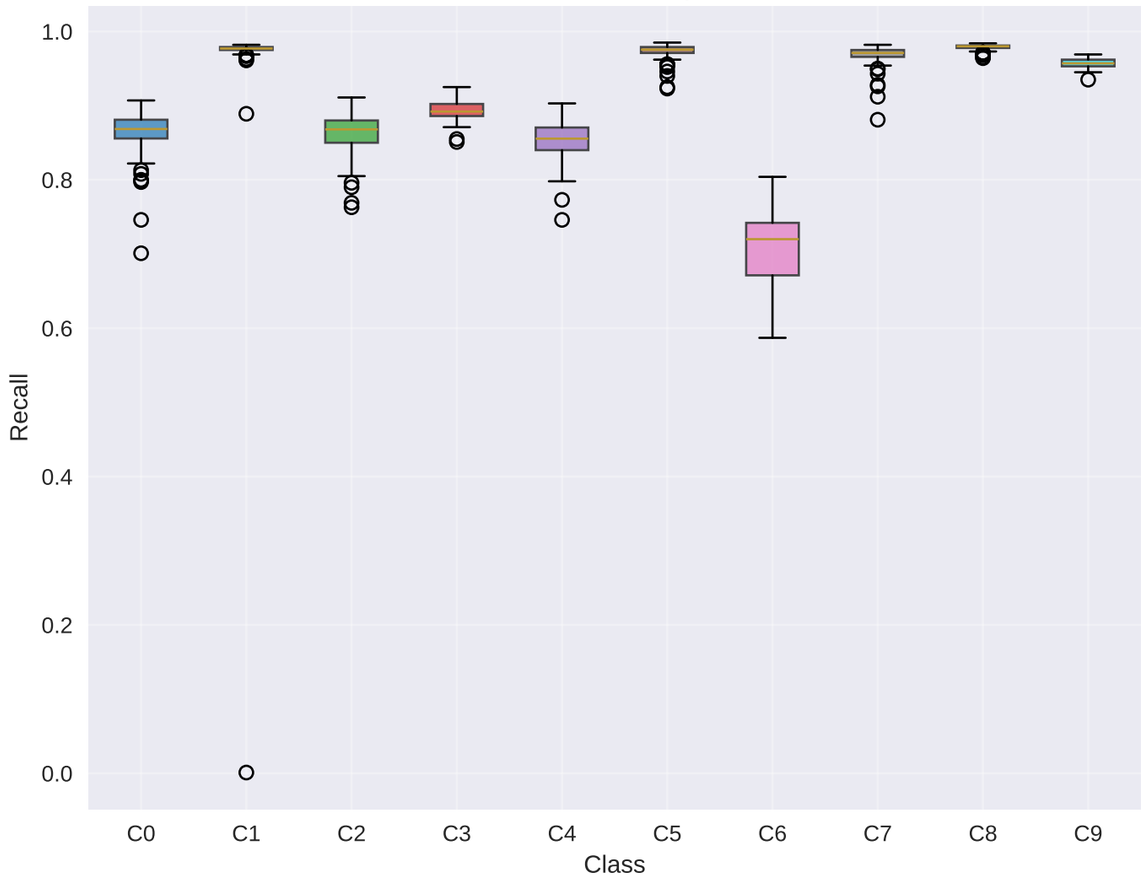


Per-Class Recall Analysis



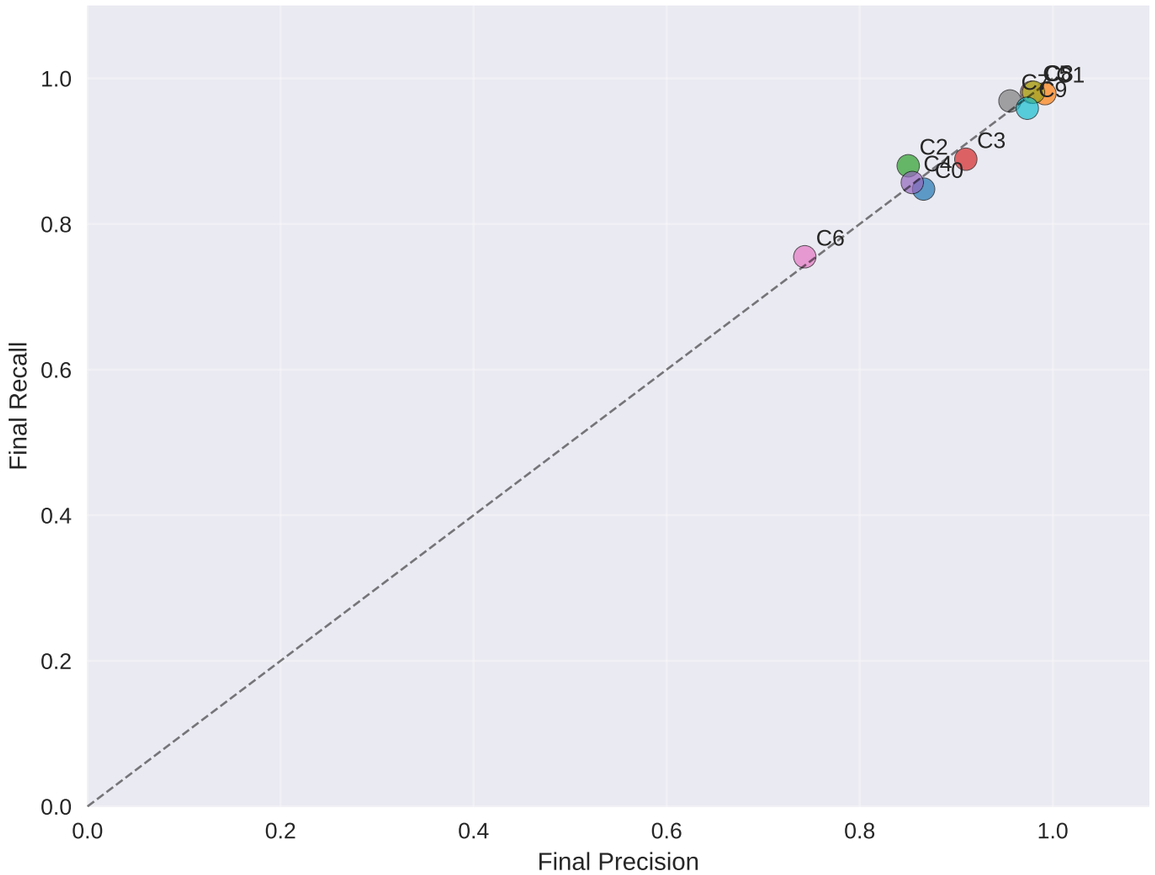




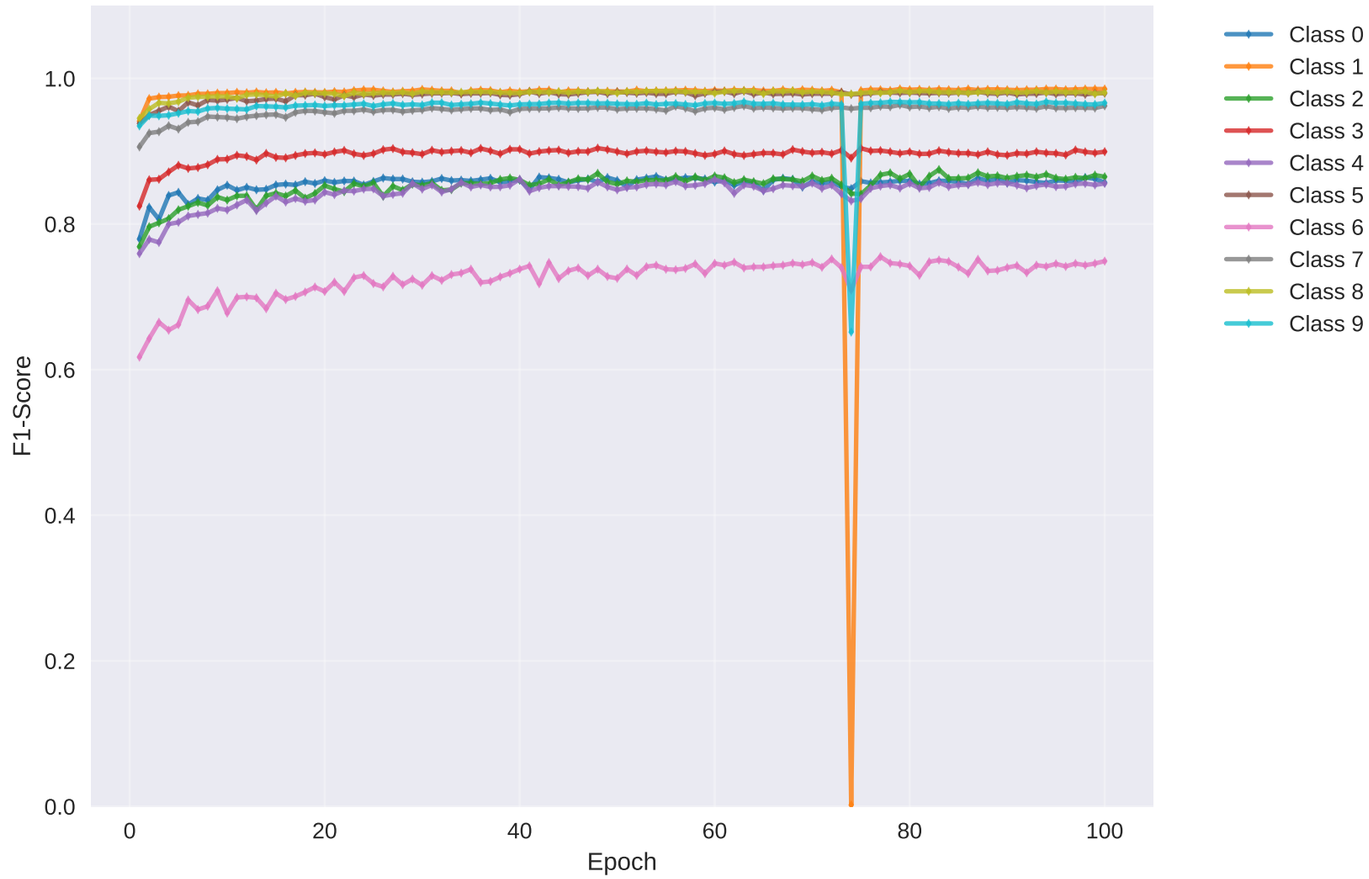


Comparative Analysis

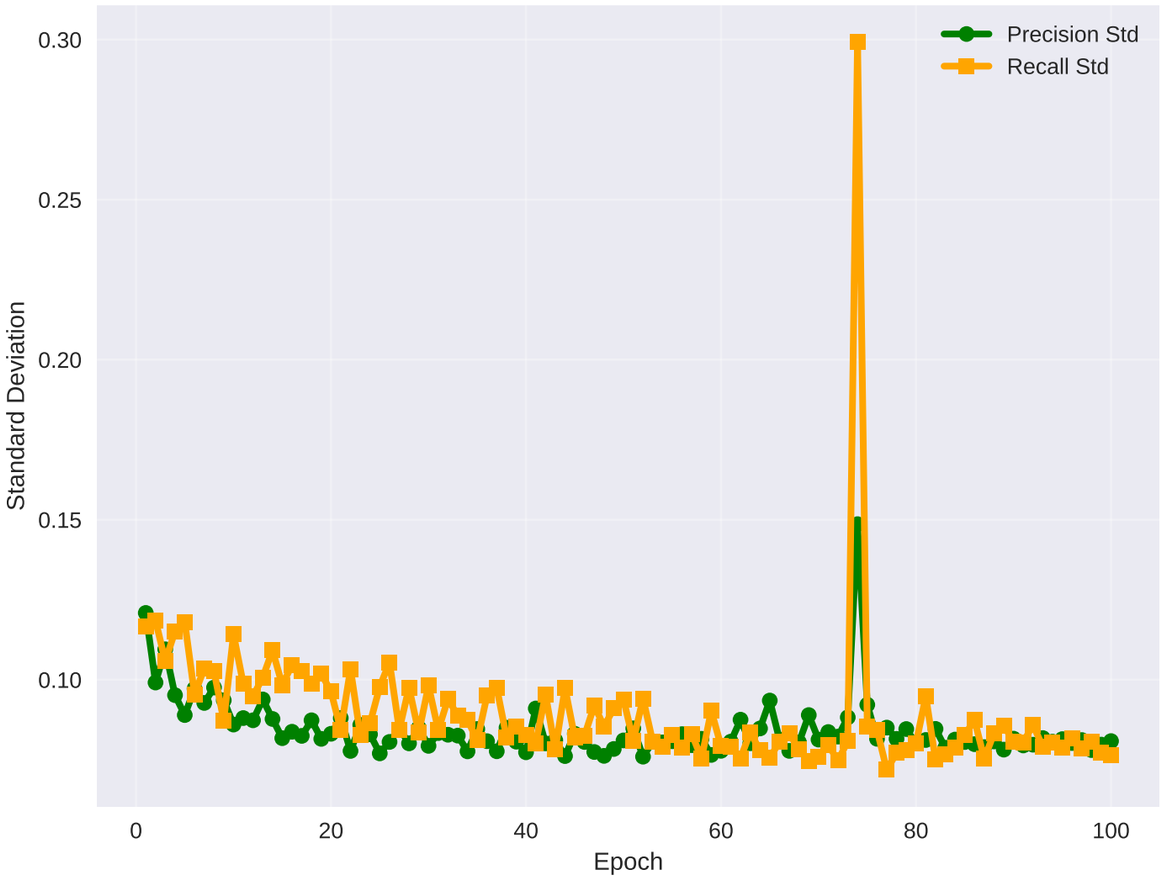
□ Precision vs Recall (Final Values)



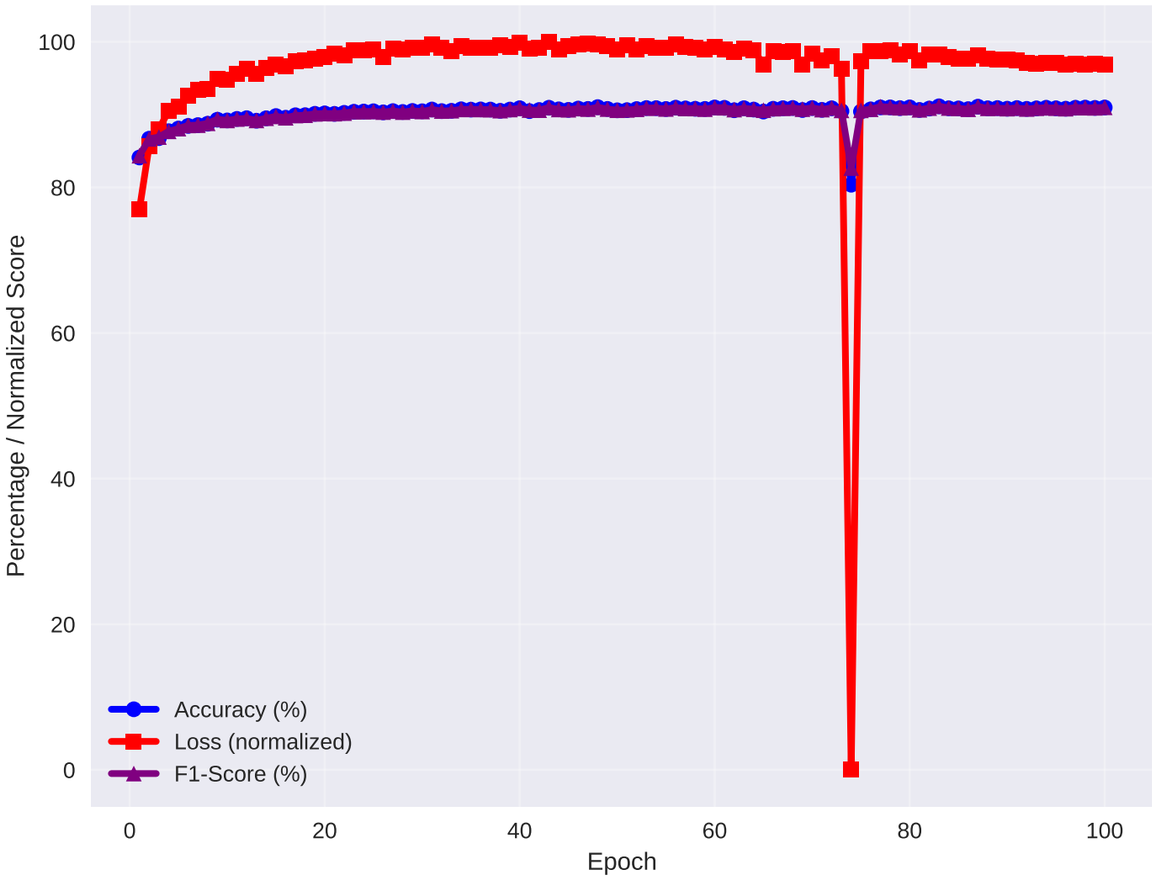
□ Per-Class F1-Score Evolution



□ Inter-Class Variance Evolution



□ Learning Curves Comparison



Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-09-01 17:51:17
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 90.98%
• Loss: 2.8395
• Average Precision: 0.910
• Average Recall: 0.910
• Average F1-Score: 0.910

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +6.87%
• Riduzione Loss: +1.6705
• Stabilità Accuracy (CV): 0.0161

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.992)
• Peggiore Precision: Classe 6 (0.743)
• Migliore Recall: Classe 5 (0.981)
• Peggiore Recall: Classe 6 (0.755)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.866, R=0.848, F1=0.857
• Classe 1: P=0.992, R=0.979, F1=0.985
• Classe 2: P=0.850, R=0.880, F1=0.865
• Classe 3: P=0.910, R=0.889, F1=0.899
• Classe 4: P=0.854, R=0.857, F1=0.856
• Classe 5: P=0.978, R=0.981, F1=0.980
• Classe 6: P=0.743, R=0.755, F1=0.749
• Classe 7: P=0.956, R=0.969, F1=0.962
• Classe 8: P=0.980, R=0.981, F1=0.981
• Classe 9: P=0.974, R=0.959, F1=0.966

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
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